Feature ID: F-6.5.16

Feature Name: Founder Operational Control Layer (FOCL)

Phase: Phase 1

---

1. User Problem Solved

The platform enforces business rules transactionally at the database layer. Automation, downgrade logic, event logging, and fallback behavior operate independently of the Founder.

However, the Founder requires:

• Clear visibility into system state
• Safe ability to approve reviews
• Controlled ability to override certain states
• Deployment visibility
• Load awareness
• Proprietary-safe investor response drafting
• Aggregated geographic awareness

Without:

• Reading raw logs
• Accessing technical dashboards
• Breaking authority boundaries
• Introducing operational risk

The Founder Operational Control Layer centralizes this capability.

---

2. Purpose

To provide a calm, secure, role-bound operational cockpit that allows the Founder to:

• Monitor system integrity
• Review blocked actions
• View load state
• Understand feature readiness
• Respond to investor inquiries
• Audit system changes

Without duplicating business rules or weakening automated protections.

This feature does not replace database-level enforcement.

It sits above it as a supervisory layer.

---

3. Subcomponents

The following subcomponents are canonical parts of F-6.5.16:

F-6.5.16.a — Founder Dashboard

Purpose:
Single-surface situational awareness.

Displays:

• Operational Mode (Normal / Elevated / Protective / Recovery)
• Connector Health Summary
• Feature Readiness Summary
• Recent Changes (last 24h)
• Pending Decision Queue items
• Regional Activity Snapshot

Design Principle:
The Founder must understand platform state within 60 seconds.

Logged Events:
founder_dashboard_viewed

F-6.5.16.b — Decision Queue

Purpose:
Surface items blocked by enforcement rules requiring manual review.

Sources include:

• Rule A (Capture → Action blocked without evidence)
• Rule B (ManualOnly → Enabled blocked without manual_review_approved)
• Connector re-enable pending QA
• Feature override request

Displays for each item:

• What happened
• Why it happened
• Risk prevented
• User impact
• Suggested action

Actions allowed:

Phone:
Approve review
Request evidence

Laptop:
Re-enable connector
Override feature state (if allowed)

All actions logged.

WAYFARER PLATFORM — PRODUCT BIBLE

F-6.5.16.b AMENDMENT

DECISION QUEUE — ACTION INBOX DOCTRINE

Version 1.0 · March 2026 · Binding

STATUS

The four action_inbox tables (action_inbox_items, action_inbox_notes, action_inbox_state_changes, action_inbox_projector_state) and all RPCs are fully built in the codebase. Developer reference documentation exists at lib/ACTION_INBOX_README.md. This amendment provides the formal product doctrine governing this system — its purpose, mutation rules, role permissions, surge behavior, and the Decision Queue UI specification. Nothing is changed in the database. This is doctrine for what already exists.

F-6.5.16.b — Decision Queue Purpose

The Decision Queue is the primary surface through which the Founder maintains awareness of platform events that require human judgment, acknowledgment, or action. It is not a task manager. It is not a support ticketing system. It is a governed, ledger-derived queue of items that the platform's automatic systems could not resolve on their own — or chose not to, by design.

Every item in the Decision Queue was created because something happened on the platform that has a consequence the Founder should know about. The item tells a story: what happened, why it matters, what was done automatically, and what (if anything) the Founder needs to do.

DOCTRINE — THE QUEUE IS INFORMATIONAL FIRST

Most items in the Decision Queue require no action. They are records of things the system handled correctly and wants the Founder to be aware of. The Founder should be able to look at the queue and feel informed, not overwhelmed. Items that genuinely require action are visually distinct. The default is: read and acknowledge.

Item Types and What Each Means

item_type	What It Is	Typical Source	Typical Priority	Founder Action
task	Something the system cannot complete without human decision. Blocked transition. Manual review required.	Feature blocked by ManualOnly rule. Auto-rollback fired but manual review flagged. Connector re-enable pending QA.	high or critical	Must act to unblock. Options presented: Approve / Deny / Investigate.

alert	Something happened that is outside normal parameters. System may have responded automatically.	Ledger integrity failure. Backup test failed. Health check degraded. Security event.	critical or high	Read and acknowledge. If auto-response was taken, verify it was correct. Decision Queue shows what the system did.
notification	Something happened that is within normal parameters but worth knowing.	Feature activated. Rollout percentage changed. New trip unlock. New user crossed a threshold.	medium or low	Read. No action required. Dismiss when done.
escalation	A condition that has persisted or worsened beyond a threshold.	Auto-rollback fired twice in 24h. Founder unreachable for 48h during critical incident (Section 8.8.9). Delegate notified.	critical	Must act. Time-sensitive. System is in a sustained non-normal state.

Item Status Lifecycle

Status	Meaning	Who Can Set	Notes
open	Item exists and has not been acted on	System (projector)	Default on creation. Appears in default queue view.
snoozed	Founder has seen it and chosen to revisit later	Founder (via snooze_action_inbox_item RPC)	snoozed_until timestamp required. Item disappears from default view until wake time. Wake is automatic.
assigned	Item has been assigned to a specific person for action	Founder (via assign_action_inbox_item RPC)	assigned_to UUID required. Used when Founder delegates review. Assigned items appear in assignee's filtered view.
resolved	Item has been addressed and is closed	Founder or System (via set_action_inbox_status RPC)	resolved_at timestamp set. Item removed from active queue. Visible in audit history. Cannot be un-resolved.

dismissed	Founder has acknowledged and closed without formal resolution	Founder (via set_action_inbox_status RPC)	Used for notifications that don't require action. Treated as resolved for queue purposes.

INVARIANT — NO DELETION

Items are never deleted. resolved and dismissed items move out of the active queue view but remain in the database permanently. The Decision Queue is an append-only record of everything the platform surfaced for Founder attention. This is an audit requirement.

Priority Definitions

Priority	Definition	Display	Response Expectation
critical	Platform is in a non-normal state that may be affecting users or the integrity of data right now. Requires immediate attention.	Red badge. Push notification sent per Section 8.8.10. Appears at top of queue regardless of time.	Acknowledge within 15 minutes during waking hours. System escalates if unacknowledged per 8.8.9.
high	Something important happened that needs Founder attention today. Not actively harming users but cannot be left indefinitely.	Amber badge. No push notification unless Founder has configured high-priority push.	Review within same business day.
medium	Routine operational event worth knowing. No urgency.	Standard badge.	Review within 48 hours.
low	Informational. No action likely needed.	Dimmed badge. Collapsed by default in queue view.	Review at Founder's convenience.

Projector — How Items Are Created

No item enters the Decision Queue directly. Every item is created by the run_action_inbox_projector() RPC, which runs as a background job on a defined schedule. The projector reads new event_ledger events since the last high-water mark, evaluates each event against projection rules, and creates inbox items for events that qualify.

Projection Rules (what events become queue items)

Event Type	item_type Created	Priority	Title Template
feature_auto_rollback_triggered	alert	high	Feature [feature_id] was automatically rolled back
ledger_integrity_failure	alert	critical	Ledger integrity failure detected — [anchor_id]
backup_restore_test_failed	alert	high	Backup restore test failed — immediate review required
region_failover_triggered	alert	critical	Regional failover triggered — [region_id]
connector_state_changed (→ ManualOnly)	task	high	Connector [connector_id] requires manual review before re-enabling
breach_founder_unreachable	escalation	critical	Founder unreachable for 48h — delegates notified
scan_credit_insufficient (threshold)	notification	medium	Deep Scan credits running low for trip [trip_id]
feature_activation_changed	notification	low	Feature [feature_id] activation changed by [actor_id]
pii_erasure_requested	task	high	PII erasure request for account [account_id] — review required
claim_filing_window_expired	notification	medium	Filing window expired for claim [claim_id]

deep_scan_credit_consumed (batch threshold)	notification	low	Deep Scan credit consumption above normal — [trip_id]

This table is not exhaustive. The projector evaluates all event types registered in event_type_registry with severity_class = warning or critical, plus specific info events listed above. New projection rules are added when new event types are registered that should surface to the Founder.

Notes System

Any authenticated user can add a note to an inbox item via the add_action_inbox_note() RPC. Notes are append-only — they cannot be edited or deleted. Notes are threaded per item and displayed chronologically in the Decision Queue item detail view.

- Use notes to record reasoning: 'Reviewed rollback — cause was a spike in error_rate due to deployment. Safe to re-enable.'
- Use notes to document actions taken outside the platform: 'Called hosting provider. Confirmed AZ issue resolved at 14:32 UTC.'
- System notes are added automatically when certain RPCs run — e.g., when auto-rollback fires, a note is added: 'Auto-rollback triggered by ERROR_RATE rule threshold exceeded (8.2% > 5.0% threshold, 90-minute window).'

Role Visibility and Permissions

Role	Read	Create Items	Change Status	Add Notes	Assign	Export
Founder	✅ All items	⛔ Only via projector	✅ All transitions	✅	✅	✅
System (projector)	✅	✅ Via run_action_inbox_projector only	⛔ Automatic transitions only (e.g. resolved when auto-rollback completes)	✅ System notes only	⛔	⛔
Support	✅ Items not marked	⛔	⛔ Can mark resolved with reason_code =	✅	⛔	⛔

	confidential		support_reviewed			
Traveler	►►	►►	►►	►►	►►	►►

Surge and Operational Mode Behavior

Operational Mode	Queue Behavior	Projector Behavior	Founder Action Available
NORMAL	Full queue visible. All actions available. Snooze, assign, resolve all enabled.	Runs on schedule. Normal batch size.	All
ELEVATED	Full queue visible. High and critical items highlighted. Snooze disabled for critical items.	Runs on shortened schedule. Batch size increased.	All — step-up verification required for structural actions
PROTECTIVE	Queue visible. Read-only except for critical item acknowledgment.	Continues running. Items accumulate. Founder notified immediately for critical items.	Acknowledge critical items. Resolve. Add notes. Cannot snooze or assign during PROTECTIVE.
RECOVERY	Queue visible. Read-only.	Paused during RECOVERY. High-water mark preserved. Resumes when NORMAL restored.	Read only. No status changes during RECOVERY.

Decision Queue UI Specification (Surface FOCL_ACTION_INBOX)

The Decision Queue UI is registered as screen surface FOCL_ACTION_INBOX at route /focl/inbox. This surface is not yet built. The following specification governs its implementation.

Default View

- Shows: all open and snoozed (past wake time) items, sorted by priority DESC then created_at ASC
- Grouped by priority: critical items at top with red section header, high below with amber, medium and low collapsed
- Each row: priority badge, item_type icon, title, feature_id tag, age ('2h ago'), next_step_hint preview
- Tapping/clicking a row expands the full item detail inline

Item Detail View

- Title (large)
- Body — plain English explanation of what happened
- What was done automatically (if anything) — pulled from metadata
- Next step hint — pre-computed guidance
- Linked event — link to event in Event Log Viewer
- Linked incident — link to incident if incident_id is set
- Notes thread — chronological, append-only
- Actions: Resolve / Snooze / Assign / Add Note / (Approve / Deny for task type)

Filter Controls

- Status filter: open | snoozed | resolved | dismissed | all
- Priority filter: critical | high | medium | low | all
- Feature filter: dropdown of feature_registry feature_ids
- Type filter: task | alert | notification | escalation | all
- Date range: last 24h | 7d | 30d | all

Phone Behavior (per Section 7.3.3 and Section 7.9.6)

- Full queue visible on phone
- Resolve and Add Note available on phone
- Assign available on phone
- Snooze available on phone for high/medium/low. Not available on phone for critical items.
- Export not available on phone

Behavioral Invariants

ID	Rule

I-01	Every item in the Decision Queue was created by run_action_inbox_projector. No authenticated session may INSERT into action_inbox_items directly.
I-02	Items are never deleted. resolved and dismissed items remain in the database. The queue is an append-only audit record.
I-03	Every status transition is recorded in action_inbox_state_changes. No status change is silent.
I-04	Critical items cannot be snoozed from a phone. The Founder must acknowledge critical items on a device capable of taking the required action.
I-05	The projector uses idempotency_key to prevent duplicate items. The same ledger event cannot produce more than one inbox item.
I-06	The projector high-water mark is stored in action_inbox_projector_state. If the projector crashes, it resumes from the last processed event — no events are skipped.
I-07	System notes are written automatically when auto-rollback fires, when containment actions execute, and when the projector processes a batch. These notes are always present before the Founder reviews the item.
I-08	The Decision Queue is always accessible in PROTECTIVE mode (read + critical acknowledge). It is never blocked. The Founder always has awareness of platform state.

F-6.5.16.b Amendment — Decision Queue Action Inbox Doctrine · Version 1.0 · March 2026

Tables built · RPCs built · Developer README complete · UI not yet built · This document is the governing doctrine

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F-6.5.16.c — Feature Readiness Panel

Purpose:
Display canonical state of all Feature IDs.

States:

• Ready
• In Development

• Disabled
• Auto-Suppressed
• Degraded Under Load

Displays:

• Feature ID
• Feature Name
• What it controls (1-line description)
• Risk Level
• Last Change Source (Manual / System / Deployment / Spike Protection)
• Evidence of State (linked event types)

Founder cannot edit core business rules from this panel.

F-6.5.16.d — System Load Awareness

Purpose:
Expose platform stress level in calm language.

Modes:

• Normal
• Elevated Load
• Protective Mode
• Recovery Mode

Displays:

• Protections activated
• Features temporarily restricted
• Connectors downgraded
• Automation paused

Must explain impact on:

• User experience
• Data integrity
• Financial exposure

No raw metrics shown by default.

F-6.5.16.e — Activity Map & Regional Load Visualization

(Previously defined; now canonically part of FOCL.)

Purpose:
Geographic aggregation of incident volume and stress patterns.

Shows:

• Incident density
• Downgrade clusters
• Manual review concentration
• Causality cluster type (Weather, ATC, Operational) — via CCO

Section 3.4 Causality Model (Dr…

Privacy:
Aggregated only.
No PII.

F-6.5.16.f — Airline Connections

Purpose:
Show health of airline automation and reply capture.

Displays per airline:

• Auto-Submission (Active / Paused)
• Reply Capture (On / Off)
• Current State (Enabled / Degraded / Manual Only / Under Review)
• Plain-language reason

Derived from downgrade logic

Section 12.5 — Logging, Observa…

Technical details hidden behind expandable view.

F-6.5.16.g — What Changed

Purpose:
Human-readable deployment and configuration summary.

Displays:

• Feature name updated
• Type of change (Code / Config / Threshold)
• Importance level
• User impact summary
• Post-change auto-protection triggers (if any)

Technical log accessible but hidden by default.

F-6.5.16.h — Ask Panel (Investor / Advisor Interface)

Purpose:
Translate inbound requests into structured, risk-scanned responses.

Capabilities:

• Feature ID mapping
• Overlap detection
• Proprietary exposure flagging
• Draft modes:

- Summary
- Technical
- Redacted
- Board Mode

Tracks:

• What was sent
• To whom
• When

No raw logs auto-exposed.

F-6.5.16.i — Explain Why Engine

Purpose:
Translate structured system events into calm explanations.

For every state change, generates:

1. What happened
2. Why
3. Risk prevented
4. User impact
5. Suggested safe action

Consumes:

• Downgrade events

Section 12.5 — Logging, Observa…

• CCO classifications

Section 3.4 Causality Model (Dr…

• Feature state changes
• Deployment triggers

No fear language.
No blame language.
No speculation.

F-6.5.16.j — Allocation Capacity Controls (Founder)

Purpose
Give the Founder a safe, calm way to temporarily add capacity to keep onboarding and core guidance stable during regional surges.

Founder-facing language
This panel does not show raw infrastructure metrics by default. It translates load into simple bands:

- Normal
- Higher than normal
- Unusually high
- Extremely high

What this control can do
Each action requires:

- Region scope (map-selected)
- Duration (default + custom hours allowed)
- Confirmation (step-up verification for high-cost actions)

Available actions:

- Add temporary capacity (general)
- Boost AI processing (faster guidance + summarization throughput)
- Boost database capacity (faster saving + retrieval of incidents, evidence, and timelines)

Why it matters to users (plain language)

- Faster onboarding completion
- Fewer stalled steps during evidence upload
- Shorter waits for guidance screens and summaries
- Less “nothing is moving” frustration during disruptions

Why it matters financially (Founder view)

- Protects conversions during high demand
- Protects retention for existing subscribers (service quality during crisis)
- Reduces drop-off from stalled onboarding flows
- Enables “stability preservation” pricing rules (see Crisis Enrollment System)

Default financial framing shown in the control

- Baseline cost (last 7 days)

- Current cost trend (today)
- Expected added cost (if you apply the selected boost)
- Predicted effect (plain language + optional deeper view)

Logging requirement
Every allocation action is written to the Founder Action Ledger with:

- who approved
- region scope
- duration
- predicted effect snapshot
- actual post-action outcome snapshot (email summary option)

F-6.5.16.k — Finance & Regional P&L (Founder)

Purpose
Give the Founder a region-aware financial dashboard that ties operational decisions to costs and revenue.

Default views

- Daily
- Weekly
- Monthly
- 13-week view (planning horizon)

Region interaction
From the Activity Map, selecting a region shows:

- Current revenue (region)

- Onboarding rate (region)
- Estimated lost revenue if onboarding slows (region)
- Cost baseline (region)
- Current cost (region)
- Predicted cost impact if allocation controls are applied

Exports
Exports are available with standardized “recipient profiles”:

- Investor
- Auditor
- Bank / lender
- Internal ops

Default export mode: **Redacted**, with recipient profile controlling what is included.

Predictive modeling activation
Financial prediction features activate only after the region has at least **30 days** of operational history.

F-6.5.16.l — Founder Action Ledger (Founder)

Purpose
A single timeline of Founder actions and major system-protection state changes, designed for auditability and post-mortems.

What it records

- Founder approvals (queue bulk actions, connector holds, recovery tests)
- Allocation Capacity Controls actions
- Crisis Enrollment policy changes (enable/disable/threshold changes)

- Major system state shifts (Normal → Protective → Recovery)

How it displays

- Calm, plain language event titles
- Click to expand: “what changed / why / user impact in one sentence”

Why it exists

- Trust and compliance
- Internal accountability
- “What happened and why” clarity during public criticism or audits

---

F-6.5.16.m — Crisis Enrollment System (Founder)

Purpose
Keep onboarding available during major disruptions without collapsing service quality for existing users.

Core rule

- Never block account creation
- Never block plan selection
- Never block evidence capture
- When strain is unusually high, onboarding may require one of these paths:

Path A: On-demand stability fee (region-scaled)
Polite, transparent framing:
“You’re joining during unusually high disruption volume in your area. This helps us keep onboarding stable for everyone during major disruptions.”

Path B: Reduced fee via verified contribution (anti-gaming rules apply)
Examples of qualifying contribution (high-quality verified evidence):

- Clear departure board photo with timestamp + gate context
- Airline app error screen with visible flight details
- Boarding pass + delay/cancel notice pairing
  System must verify uniqueness and quality before credit is applied.

Path C: Waitlist onboarding (no fee)
User provides minimal info and is queued. When regional capacity returns, user is notified to finish onboarding without fee.

Founder controls

- Enable/disable crisis enrollment for a region
- Set thresholds (by region, auto-adjusted by region maturity)
- Define fee bands (region-scaled)
- Define contribution credit rules (anti-abuse)

Logging requirement
All crisis enrollment policy changes are written to the Founder Action Ledger.

F-6.5.16.n — Crisis Communication Drafting (Founder)

Purpose
Help the Founder produce accurate, non-technical, non-alarmist explanations during incidents using internal evidence.

Inputs (source-bound)

- Event log summaries (read-only)
- What changed feed
- Current operational mode + affected surfaces
- Region map highlights

Outputs

- Founder internal note
- User-facing notice copy
- Public transparency update draft
  All drafts must include a one-sentence “what changes for users” summary.

F-6.5.16.o

Interpretive Diagnostics & Drift Observatory

(Subcomponent of Founder Operational Control Layer)

1. Purpose

The Interpretive Diagnostics & Drift Observatory (“IDDO”) is a supervisory integrity system that monitors long-term interpretive behavior of the AI layer.

Its function is to detect sustained deviations in interpretive confidence patterns relative to established structural baselines.

It does not:

- Modify user outputs
- Auto-correct interpretations
- Adjust confidence labels
- Change clauses
- Alter routing

It observes, correlates, and escalates only when necessary.

---

2. Core Doctrine

Drift is defined as:

Sustained deviation of interpretive classification patterns beyond established tolerance thresholds relative to baseline.

Drift is:

- A signal
- Not a verdict
- Not an error by default
- Not an immediate failure condition

Drift may reflect:

- Policy language evolution
- Carrier mix change
- Parsing updates
- Clause extraction updates
- Model revision
- Loosening
- Tightening
- Bias
- Improvement

Drift must always be explainable.

Unexplained drift requires review.

---

3. Calibration Phase (Baseline Establishment)

3.1 Calibration Window

Baseline is established after:

- First 90 days of stable operation
  OR
- First 1,000 completed interpretive cases
  (whichever occurs later)

During calibration:

- No alerts issued
- Metrics recorded only
- Confidence distributions logged daily

At completion:

- Baseline confidence distribution frozen
- Founder Step-Up verification required
- Baseline snapshot versioned
- Logged under Section 8.4 governance ledger

---

4. Baseline Architecture

Hybrid baseline model is adopted.

4.1 Frozen Baseline

Locked after calibration.

Used as doctrinal reference anchor.

4.2 Rolling Baseline

30-day rolling average updated continuously.

Used to detect rapid short-term anomalies.

---

5. Drift Threshold Rules

5.1 Individual Confidence Enum Monitoring

For each confidence label defined in Section 9.2:

Alert condition:

Absolute deviation ≥ 10% from frozen baseline
AND
Minimum 250 cases in rolling window
AND
Sustained ≥ 7 consecutive days

5.2 Overall Distribution Skew Monitoring

Alert condition:

Total variation divergence ≥ 10% from frozen distribution
AND sustained ≥ 7 days

---

6. Drift Severity Levels

Drift severity is internal, not user-facing.

Level 0 — Monitor (Default)

Continuous background tracking.
No surface display.

Level 1 — Advisory

Deviation detected but:

- Duration < 7 days
  OR

- Sample size marginal

System records internally.
No founder notification.

Level 2 — Requires Review

Deviation meets all conditions:

- ≥10% deviation
- ≥250 cases
- ≥7 days sustained
- Correlation check complete

At this stage:

1. System auto-correlates with:
    - Policy version hash changes
    - Clause extraction updates
    - Parsing updates
    - Model version changes
    - Feature activation logs (8.7)
    - Stress test history (Section 15)
2. Generates structured Drift Review Dossier.
3. Harden parameters automatically:
    - Freeze baseline adaptation
    - Lock confidence thresholds
    - Pause rolling baseline recalibration

4. Notify Founder.

---

7. Founder Notification Protocol

Founder is notified only at Level 2.

Notification includes:

Title:
Confidence Pattern Shift Detected

Contents:

- Metric affected
- Frozen baseline %
- Current rolling %
- Deviation %
- Duration sustained
- Case volume
- Correlated system changes
- Recommended review actions

No panic language.
No urgency framing.
No emotional signaling.

---

8. Parameter Hardening During Review

When Level 2 triggered:

System automatically:

- Freezes rolling baseline updates
- Locks enum threshold tolerance
- Suspends baseline recalibration
- Prevents silent normalization

This protects doctrine during investigation.

Step-Up required to:

- Accept drift as new normal
- Reset baseline
- Modify threshold tolerance
- Override severity state

All actions logged under 8.4.

---

9. Relationship to Section 15 (Stress Testing)

Upon Level 2 trigger:

System recommends:

Targeted stress scenarios relevant to drift metric.

Stress execution remains separate authority.

Drift detection does not auto-run stress battery.

---

10. Policy Version Drift Protection

Every uploaded policy document is:

- Parsed
- Canonicalized
- Clause segmented
- Hashed (policy_hash_id)

If same product name yields different hash:

Emit:
policy_version_mismatch_detected

Document drift is tracked separately from interpretive drift.

Correlation engine checks document drift before escalating interpretive drift.

---

11. Founder Cockpit Surface

Panel name:

Interpretive Stability

Displayed only when:

- Advisory exists
  OR
- Review required

Default state: hidden.

When visible, shows:

- Metric
- Baseline
- Current

- Deviation
- Duration
- Volume
- Correlated change history

No toggles exposed.

Actions require step-up.

---

12. Survivability Doctrine

Platform must continue operating normally during drift state.

Drift does not:

- Block interpretations
- Change confidence labels
- Halt routing
- Affect user interface

It is supervisory only.

---

4. Authority Boundary Matrix

The Founder Operational Control Layer must enforce explicit authority boundaries.

Irreversible or structural actions require:

• Laptop access
• Step-up verification
• Logged justification

System-protected states that cannot be overridden:

• Event log immutability
• Rule A enforcement
• Rule B enforcement
• Database-level transactional rules

This prevents “Founder override everything.”

---

5. Phone vs Laptop Policy

Phone:
• Read-only by default
• Approve low-risk manual reviews
• View system state
• Draft Ask responses

Laptop:
• Change feature state
• Re-enable automation
• Adjust thresholds
• Lock feature from auto-reenable
• Enter Surge Review Mode

Step-up verification required for:

• Threshold changes
• Re-enabling auto-suppressed features
• Manual override of protective mode

---

6. Logging & Audit

Every Founder action generates event_logs entry:

• actor_id
• action_type
• timestamp

• justification (if required)
• previous_state
• resulting_state

No update/delete.
Insert-only.

---

7. Failure Modes & Safe Defaults

If Founder portal fails:

• Core automation unaffected.
• Manual fallback unaffected.
• Database rules unaffected.

Founder cockpit must be supervisory only.

Platform must survive 24–48 hours without Founder login.

F-6.5.16 Amendment — FOCL Feature Intelligence Layer

Version 1.1 Amendment | March 2026
Supersedes: F-6.5.16.c (Feature Readiness Panel) — replace entirely
Adds: F-6.5.16.j (Feature Intelligence Panel), feature_registry schema extensions, M-22 migration

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Why This Amendment Exists

The current Feature Readiness Panel (F-6.5.16.c) shows five states: Ready, In Development, Disabled, Auto-Suppressed, Degraded Under Load. It shows feature ID, name, description, risk level, last change source, and linked event types.

That is a status board. What the Founder needs is an intelligence panel — one that shows not just what state a feature is in, but what phase it's at, whether it has sub-capabilities waiting to be enabled, whether it's healthy or degrading, what's happening to it right now, and what action (if any) is required. Every feature that exists — not just the ones that are enabled — needs to be visible, togglable, and monitored.

The doctrine is: **if it's a feature, it has a toggle. If it has a toggle, it has health monitoring. If it has health monitoring, it has auto-rollback rules. If something goes wrong, the Founder sees it in plain language, and the platform tries to fix it before the Founder has to act.**

F-6.5.16.j — Feature Intelligence Panel (replaces F-6.5.16.c)

Purpose

Single-surface visibility into every feature on the platform — its current state, its capability tier, its health, its rollout status, and any active issues — with the ability to act directly from the panel without navigating elsewhere.

The Founder must be able to answer the following questions within 30 seconds of opening this panel:

- What features are fully live?
- What features are partially live (MVP tier, Phase 2 waiting)?
- What features are off and why?
- What features are currently degrading or in auto-rollback?
- What features have sub-capabilities that can be independently toggled?
- Is anything broken right now that needs my attention?

Per-Feature Row Display

Every row in the panel shows:

Field	What It Shows
Feature ID	e.g., F-6.5.8
Feature Name	Plain English name
Phase	MVP · Phase 2 · Phase 2+
Capability Tier	1 = base · 2 = extended · 3 = full (see Section below)

Status	LIVE · PARTIAL · DISABLED · SUPPRESSED · DEGRADED · ROLLED_BACK
Rollout %	0–100% — what percentage of eligible scope receives this feature
Health	HEALTHY · DEGRADED · UNHEALTHY
Active Issues	Count of open Decision Queue items linked to this feature
Last Changed	Timestamp + change source (Manual / System / Auto-Rollback / Deployment)
Sub-Capabilities	Expandable — shows any F-X.X.X-SUFFIX entries (e.g., F-6.5.8-LIVE)
Pending Extension	Flag if a Phase 2 sub-capability exists but is not yet enabled

Status Definitions

Status	Meaning
LIVE	Enabled, rollout % > 0, health HEALTHY
PARTIAL	Enabled at MVP tier, Phase 2 sub-capability exists but disabled
DISABLED	Manually disabled by Founder or never activated
SUPPRESSED	Auto-disabled by governance guard or surge mode
DEGRADED	Enabled but health shows DEGRADED — auto-rollback rules may be evaluating
ROLLED_BACK	Auto-rollback fired — feature was reduced to lower rollout % or disabled

Capability Tier Model

Every feature gets a capability tier. This tells the Founder what level of the feature is currently active.

Tier 1 — Base: Core MVP functionality. No external connectors required.
Tier 2 — Extended: Adds integrations, richer data, or cross-feature connections. May require connectors.
Tier 3 — Full: Complete feature including Phase 2 live data, all connectors active, all sub-capabilities enabled.

Example: F-6.5.8 Active Disruption Options Engine

- Tier 1: Coverage-aware guidance, preference capture, evidence parallel capture (no connector)
- Tier 2: Historical fare range estimates (lightweight external data)
- Tier 3: Live GDS flight search, live booking links (F-6.5.8-LIVE connector active)

The panel shows which tier is currently active and what would be needed to advance to the next tier (missing connector, pending licensing, Phase 2 migration required).

Toggle Controls

Every feature row has inline controls:

Enable / Disable toggle — Founder can flip on/off. Requires step-up confirmation for features that are currently serving live users. All toggles emit `feature_activation_changed` to the event ledger.

Rollout % slider — Set 0–100 for gradual rollout. Deterministic hash ensures the same users get the same experience on every session. Can be set while users are actively using the platform — in-flight sessions are not interrupted, the new gate applies to new requests only.

Sub-capability toggles — If a feature has sub-capabilities (e.g., F-6.5.8-LIVE), each sub-capability has its own independent toggle and rollout % visible in the expanded row.

Auto-rollback rules — Tap to view/edit the rollback rules for this feature. Rules define: metric type (error rate, suppression spike, battery warning count, inbox spike), threshold value, observation window, cooldown period, and rollback action (reduce to % or disable entirely).

Health Monitoring Per Feature

Every enabled feature is evaluated on a rolling basis by `evaluate_feature_rollout_health()`. The result is stored in `feature_rollout_health_state` and visible in the panel.

When health degrades:

- **DEGRADED:** Panel row turns amber. Founder sees it. No automatic action yet. The Explain Why Engine (F-6.5.16.i) generates a plain-language explanation.
- **UNHEALTHY:** Panel row turns red. Auto-rollback rules are evaluated. If a rule threshold is exceeded and cooldown has passed, `run_feature_rollout_auto_rollback()` fires.

  The Founder sees an action inbox item: what happened, what the system did, what the current state is, whether manual review is needed.

The Founder almost never has to act manually on health issues — the auto-rollback rules handle them. The panel and inbox keep the Founder informed.

Rollout Behavior During Live Usage

Rollout changes take effect for new requests only. In-flight sessions (a traveler currently mid-incident, mid-scan, mid-draft) are not interrupted. The deterministic hash is evaluated per-request, so a user who was eligible at rollout % = 50 remains eligible when the Founder bumps to 75% — their hash result doesn't change.

Rollouts can be scoped to:

- **INCIDENT** — affects per-incident feature eligibility
- **USER** — affects per-user feature eligibility (same user always gets same result)
- **TRIP** — affects per-trip feature eligibility
- **REGION** — affects all users in a geographic region

Regional rollouts (entire countries, entire regions) use the region_id field on `feature_activation_state`. A feature can be LIVE at 100% in the US and DISABLED in EU pending regulatory clearance — these are independent rows.

Platform delivery (web app, iOS, Android, tablet) does not affect feature eligibility — the gate is evaluated server-side. The client receives the result. No client-side feature flag evaluation.

The "Something Is Wrong" Experience

When a feature has an active issue, the Founder sees:

1. **Panel row highlighted** — amber (DEGRADED) or red (UNHEALTHY / ROLLED_BACK)
2. **Active Issues count** — tapping it opens the relevant Decision Queue items
3. **Explain Why summary** — one-paragraph plain English: what happened, why, what risk was prevented, what the system did automatically, what (if anything) the Founder needs to do
4. **Action buttons** — only shown when manual action is actually needed. If the auto-rollback handled it: "System auto-rolled back to 25%. No action required. Tap to review." If manual action is needed: "Manual review required. [Approve] [Investigate] [Override]"

The doctrine is: **the system handles it first. The Founder is informed, not interrupted.**

Schema Extensions Required — Migration M-22

feature_registry additions

```
ALTER TABLE feature_registry
  ADD COLUMN IF NOT EXISTS phase text NOT NULL DEFAULT 'MVP'
    CHECK (phase IN ('MVP', 'Phase2', 'Phase2Plus')),
  ADD COLUMN IF NOT EXISTS capability_tier_current int NOT NULL DEFAULT 1
    CHECK (capability_tier_current BETWEEN 1 AND 3),
  ADD COLUMN IF NOT EXISTS capability_tier_max int NOT NULL DEFAULT 1
    CHECK (capability_tier_max BETWEEN 1 AND 3),
  ADD COLUMN IF NOT EXISTS has_pending_extension boolean NOT NULL DEFAULT false,
  ADD COLUMN IF NOT EXISTS parent_feature_id text REFERENCES
feature_registry(feature_id),
  ADD COLUMN IF NOT EXISTS requires_connector text,
  ADD COLUMN IF NOT EXISTS connector_status text DEFAULT 'not_required'
    CHECK (connector_status IN
('not_required','required_unlicensed','licensed_not_integrated','active'));
```

Field explanations:

- *phase*: which release phase this feature belongs to
- *capability_tier_current*: what tier is currently active (1–3)
- *capability_tier_max*: what tier is achievable when fully built out
- *has_pending_extension*: true if a sub-capability (Phase 2 extension) exists but is disabled
- *parent_feature_id*: for sub-capabilities like F-6.5.8-LIVE, points to F-6.5.8
- *requires_connector*: plain English name of required external connector, if any
- *connector_status*: whether the connector is licensed and integrated

UPDATE statements for all existing features (include in M-22)

```
-- F-6.5.1 Policy Parsing — MVP, Tier 1/1, no connector
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1
WHERE feature_id='F-6.5.1';
```

```
-- F-6.5.2 Coverage Graph — MVP, Tier 1/2 (Tier 2 = statutory rights layer when Section 3.6 complete)
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=2, has_pending_extension=true WHERE feature_id='F-6.5.2';

-- F-6.5.3 CC Benefit Orchestration — MVP, Tier 1/2 (Tier 2 = live airline API)
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=2, has_pending_extension=true, requires_connector='Live Airline API', connector_status='required_unlicensed' WHERE feature_id='F-6.5.3';

-- F-6.5.4 Airline Disruption Intelligence — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1 WHERE feature_id='F-6.5.4';

-- F-6.5.5 Claim Routing — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1 WHERE feature_id='F-6.5.5';

-- F-6.5.6 Evidence Capture — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1 WHERE feature_id='F-6.5.6';

-- F-6.5.7-ENRICH Incident Timeline Enrichment — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1 WHERE feature_id='F-6.5.7-ENRICH';

-- F-6.5.8 Active Disruption Options Engine (UPDATE name from Causality Linking Engine)
UPDATE feature_registry SET
  display_name='Active Disruption Options Engine',
  description='Coverage-aware alternative transport guidance during active incident. Phase 2: live options via GDS connector.',
  phase='MVP', capability_tier_current=1, capability_tier_max=3,
  has_pending_extension=true, requires_connector='GDS Flight Search API', connector_status='required_unlicensed'
WHERE feature_id='F-6.5.8';

-- F-6.5.8-LIVE sub-capability (INSERT)
INSERT INTO feature_registry (feature_id, display_name, description, default_enabled, minimum_mode, phase, capability_tier_current, capability_tier_max, parent_feature_id, requires_connector, connector_status)
VALUES ('F-6.5.8-LIVE', 'Active Disruption Options — Live Search', 'Phase 2: GDS connector for live flight/transport options during active incident', false, 'NORMAL', 'Phase2', 1, 1, 'F-6.5.8', 'GDS Flight Search API', 'required_unlicensed');
```

```
-- F-6.5.9 Participant Readiness Checklist (UPDATE name from Rebooking Event Log)
UPDATE feature_registry SET
  display_name='Participant Travel Readiness & Checklist Engine',
  description='Residence-personalized travel readiness checklist for every trip participant. Auto-updates on itinerary change.',
  phase='MVP', capability_tier_current=1, capability_tier_max=2,
  has_pending_extension=true, requires_connector='Visa/Entry Requirements API', connector_status='required_unlicensed'
WHERE feature_id='F-6.5.9';

-- F-6.5.13 Deep Scan — MVP, Tier 1/3 (full tier = all axes active with licensed data sources)
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=3, has_pending_extension=true WHERE feature_id='F-6.5.13';

-- F-6.5.14 Claim Packet Generator — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1
WHERE feature_id='F-6.5.14';

-- F-6.5.15 Claim Progress Tracking — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1
WHERE feature_id='F-6.5.15';

-- F-6.5.16 FOCL — MVP, Tier 1/1
UPDATE feature_registry SET phase='MVP', capability_tier_current=1, capability_tier_max=1
WHERE feature_id='F-6.5.16';
```

INSERT statements for features not yet registered (include in M-22)

```
INSERT INTO feature_registry (feature_id, display_name, description, default_enabled, minimum_mode, phase, capability_tier_current, capability_tier_max) VALUES
  ('F-6.5.17', 'Trip Draft Engine',                  'Voice-first and manual trip itinerary drafting with readiness gate',            false, 'NORMAL', 'MVP',      1, 1),
  ('F-6.5.18', 'Route Normalization & Validation',     'Schedule conflict detection, connection buffer validation, route feasibility',   false, 'NORMAL', 'MVP',      1, 1),
  ('F-6.5.19', 'Activity Auto-Population',            'AI-suggested and creator-sourced activity recommendations for trip drafts',     false, 'NORMAL', 'MVP',      1, 2),
  ('F-6.5.20', 'Route Environmental Intelligence',     'Weather and environmental context per route segment',                           false, 'NORMAL', 'MVP',      1, 2),
  ('F-6.6.2',  'Protective Safety Mode (PSM)',         'Elevated safety posture for at-risk travelers during active incidents',         false, 'NORMAL', 'Phase2',   1, 1),
  ('F-6.6.3',  'Dual-Presence Verification (DPV)',     'Biometric + device verification for high-risk account actions',                 false, 'NORMAL', 'Phase2',   1, 1),
  ('F-6.6.4',  'Emergency Location Availability',      'Guardian-controlled location sharing during emergencies',                       false, 'NORMAL', 'Phase2',   1, 1),
```

('F-6.6.6', 'Emergency Delegate & Backup Contact', 'Designated backup contact with scoped platform access', false, 'NORMAL', 'Phase2', 1, 1),
('F-6.6.10', 'Medical Tourism Module', 'Verified provider discovery and transparent pricing for medical travel', false, 'NORMAL', 'Phase2Plus',1, 1),
('F-6.6.11', 'Creator-Indexed Travel Discovery', 'Travel content creator videos indexed for destination and activity discovery', false, 'NORMAL', 'Phase2', 1, 1),
('F-6.7.1', 'Founder Operational Command Engine', 'Global operational oversight — regional stability, automation, financials', false, 'NORMAL', 'MVP', 1, 1);

Complete Feature Registry — All Features, All Phases

Feature ID	Name	Phase	Tier Current / Max	Has Extension	Connector Required
F-6.5.1	Policy Parsing & Clause Extraction	MVP	1/1	No	None
F-6.5.2	Coverage Graph & Overlap Resolution	MVP	1/2	Yes (statutory layer)	None for Tier 1
F-6.5.3	Credit Card Benefit Orchestration	MVP	1/2	Yes	Live Airline API (Phase 2)
F-6.5.4	Airline Disruption Intelligence	MVP	1/1	No	None
F-6.5.5	Claim Routing & Sequencing Engine	MVP	1/1	No	None
F-6.5.6	Evidence Capture & Inbox Import	MVP	1/1	No	None
F-6.5.7-ENRICH	Incident Timeline Enrichment	MVP	1/1	No	None

F-6.5.8	Active Disruption Options Engine	MVP	1/3	Yes (F-6.5.8-LIVE)	GDS Flight Search (Phase 2)
F-6.5.8-LIVE	Active Disruption Options — Live Search	Phase 2	1/1	No	GDS Flight Search API
F-6.5.9	Participant Travel Readiness & Checklist	MVP	1/2	Yes	Visa/Entry API
F-6.5.10	Carrier Discrepancy Detection	MVP	1/1	No	None
F-6.5.11	Regulatory-Aware Incident Reporting	MVP	1/1	No	None (absorbed into F-6.5.13)
F-6.5.12	Authority-Driven Travel Disruptions	MVP	1/1	No	None (absorbed into F-6.5.13)
F-6.5.13	Deep Scan Intelligence Engine	MVP	1/3	Yes (axes unlock with data sources)	FlightAware/OAG, Weather APIs
F-6.5.14	Claim Packet Generator	MVP	1/1	No	None
F-6.5.15	Claim Progress Tracking & Outcome	MVP	1/1	No	None
F-6.5.16	FOCL — Founder Operational Control	MVP	1/1	No	None
F-6.5.17	Trip Draft Engine	MVP	1/1	No	None

F-6.5.18	Route Normalization & Validation	MVP	1/1	No	None
F-6.5.19	Activity Auto-Population	MVP	1/2	Yes	Creator API (Phase 2)
F-6.5.20	Route Environmental Intelligence	MVP	1/2	Yes	Premium Weather API (Phase 2)
F-6.6.1	Deferred Concern & Escalation Ladder	Phase 2+	1/1	No	None
F-6.6.2	Protective Safety Mode	Phase 2	1/1	No	None
F-6.6.3	Dual-Presence Verification	Phase 2	1/1	No	None
F-6.6.4	Emergency Location Availability	Phase 2	1/1	No	None
F-6.6.6	Emergency Delegate & Backup Contact	Phase 2	1/1	No	None
F-6.6.9	Financial Modeling & Allocation Impact	Phase 2	1/1	No	None
F-6.6.10	Medical Tourism Module	Phase 2+	1/1	No	None
F-6.6.11	Creator-Indexed Travel Discovery	Phase 2	1/1	No	None

F-6.7.1	Founder Operational Command Engine	MVP	1/1	No	None
F-CARRIER-DEEP	Carrier Portal Deep-Linking	Phase 2	1/1	No	Carrier Portal APIs
F-12.3	Data Pipelines (Landing Zone)	MVP	1/1	No	None
F-7.4	Voice-First Interaction Model	MVP	1/1	No	None

Code Items — What Still Needs to Be Built

Already in codebase (confirmed from migrations)

- `feature_registry` table ✅
- `feature_activation_state` table with rollout extensions ✅
- `feature_rollout_rules` table ✅
- `feature_rollout_health_state` table ✅
- `set_feature_activation_state()` RPC ✅
- `set_feature_rollout_percentage()` RPC ✅
- `set_feature_rollout_rules()` RPC ✅
- `evaluate_feature_rollout_health()` RPC ✅
- `run_feature_rollout_auto_rollback()` RPC ✅
- `is_feature_eligible()` deterministic hash function ✅
- FOCL screen surfaces registered (FOCL_FEATURE_CONSOLE, FOCL_FEATURE_ROLLOUT_CONTROL, FOCL_FEATURE_ROLLOUT_HEALTH, FOCL_FEATURE_ROLLOUT_RULES) ✅

Not yet built — Migration M-22 required

- *phase*, *capability_tier_current*, *capability_tier_max*, *has_pending_extension*, *parent_feature_id*, *requires_connector*, *connector_status* columns on *feature_registry*
- UPDATE statements for all existing registered features
- INSERT statements for 11 features not yet registered (F-6.5.17 through F-6.7.1)
- F-6.5.8 and F-6.5.9 display_name/description corrections
- F-6.5.8-LIVE sub-capability registration
- New event types: *feature_tier_advanced*, *feature_extension_enabled*, *feature_connector_activated*

FOCL UI — Not yet built (screen surfaces registered, no frontend)

- Feature Intelligence Panel UI (replaces Feature Readiness Panel)
- Per-feature expanded row with sub-capability toggles
- Capability tier indicator and "next tier requires" display
- Connector status display per feature
- Health status inline with amber/red highlighting
- Auto-rollback history per feature
- Rollout % slider with scope type selector (INCIDENT / USER / TRIP / REGION)
- Regional rollout map (which countries/regions have which features at which %)

Behavioral Invariants

ID	Rule
I-01	Every feature in feature_registry has a toggle. No feature exists outside the registry.
I-02	Every enabled feature with rollout_percentage > 0 has at least one auto-rollback rule. The system will not run a feature at any percentage without a health safety net.
I-03	Rollout changes apply to new requests only. In-flight sessions are never interrupted by a rollout change mid-session.

I-04	The deterministic hash is stable — the same user/incident/trip always receives the same feature gate result for a given rollout %. Changing rollout % only affects users whose hash falls in the new range.
I-05	Regional rollout is independent per region_id. A feature can be LIVE in one region and DISABLED in another simultaneously.
I-06	Sub-capabilities (e.g., F-6.5.8-LIVE) are independently togglable and have their own rollout %, health monitoring, and auto-rollback rules. Disabling a sub-capability never disables the parent feature.
I-07	When auto-rollback fires, it always creates an action inbox item. The Founder is always informed. Auto-rollback never happens silently.
I-08	The Founder can see every feature, its state, its health, and its history from a single panel. No feature is hidden from FOCL.
I-09	Platform delivery channel (web / iOS / Android / tablet) does not affect feature eligibility. All gates are server-side.
I-10	All feature state changes — whether manual, system-triggered, or auto-rollback — are emitted to the event ledger. No feature state change is silent.

F-6.5.16 Amendment v1.1 — FOCL Feature Intelligence Layer | March 2026
Supersedes F-6.5.16.c | Adds F-6.5.16.j | Requires Migration M-22
Backend infrastructure: complete in codebase | Frontend panel: not yet built | Registry: incomplete — M-22 required